Huawei Access Node JAY-ACC-13 QoS Congestion Event Root Cause Analysis and Resolution

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9595

Description

This Standard Operating Procedure (SOP) outlines the necessary steps to identify and resolve Quality of Service (QoS) congestion events on Huawei Access Node JAY-ACC-13 located at the Jayanagar Metro Site. The failure domain in question pertains to microbursts, interface tail drops, and QoS queue starvation. These issues can arise due to the node's role in managing access traffic and the vendor-specific CLI dialect used for configuration and troubleshooting. In a Huawei Access Node, QoS is implemented by the Network Service Engine (NSE), which is responsible for shaping and policing traffic according to predefined policies. When QoS policies are misconfigured or inadequate, it can lead to congestion events, manifesting as microbursts, interface tail drops, or QoS queue starvation. Furthermore, the Access Node's architecture, which includes a combination of hardware and software components, can contribute to the occurrence of these events. The NSE's processing capabilities, as well as the buffer sizes and queue depths of the node's interface cards, can all impact the node's ability to handle traffic and prevent congestion. Therefore, a comprehensive analysis of the node's configuration, as well as its hardware and software components, is essential to identifying the root cause of the congestion event and implementing the necessary corrective actions.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that all necessary equipment and personnel are available.

- Check the status of the redundant link on the Access Node to ensure it is functioning correctly.

- Gather all relevant log files and configuration data from the Access Node for analysis.

Potential Root Causes

Cause	Probability
Insufficient QoS policy configuration	High
Misconfigured NSE processing parameters	Medium
Hardware component failure or bottleneck	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos statistics interface GigabitEthernet0/0/0,
Expected Output: QoS statistics for interface GigabitEthernet0/0/0:,
Description: This command displays QoS statistics for the specified
interface, including packet count, byte count, and QoS policy hit
count. The output can help identify if QoS policies are being applied
correctly and if there are any issues with traffic shaping or
```

```
policing.

Command: display nse statistics, Expected Output: NSE statistics:,
Description: This command displays NSE statistics, including
processing capacity, packet buffer size, and queue depth. The output
can help identify if the NSE is functioning correctly and if there are
any issues with packet processing or buffer management.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: configure terminal display qos policy , Description: Verify the QoS policy configuration and adjust as necessary to ensure that it is correctly shaping and policing traffic. Use the 'display qos policy' command to view the policy configuration and make any necessary changes.

Step 2: Step: Reset NSE processing parameters, Command: reset nse processing-parameters, Description: Reset the NSE processing parameters to their default values. This can help resolve issues related to misconfigured NSE processing parameters.

Post-Check Verification

```
Command: display qos statistics interface GigabitEthernet0/0/0,
Expected Output: QoS statistics for interface GigabitEthernet0/0/0:,
Description: Verify that QoS statistics for the interface are within
expected ranges and that QoS policies are being applied correctly.

Command: display nse statistics, Expected Output: NSE statistics:,
Description: Verify that NSE statistics are within expected ranges and
that packet processing and buffer management are functioning
correctly.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo configure terminal display qos policy , Description: Restore the original QoS policy configuration to prevent any unintended changes to traffic shaping and policing.

- Step: Restore original NSE processing parameters, Command: undo reset nse processing-parameters, Description: Restore the original NSE processing parameters to prevent any unintended changes to packet processing or buffer management.